RES POLYTECHNIC, BANGALORE-91

CIE Practice Test IV - Model Answers

System and Network Administration | Course Code 25CS34I | Marks 50 | Semester III

Practical test: write the aim, procedure/commands, observation and result for each task in the observation book. (Marks: a) 10, b) 10, c) 15, d) 10, e) 5 portfolio.)

TASK a) Disk Management and System Restore (10 marks)

a1) Create and format a new partition

Aim: to create and format a new partition using Disk Management.

Procedure:

1. Press **Win + X → Disk Management** (or run `diskmgmt.msc`).
2. Right-click an **Unallocated** space (or shrink a volume first to make space).
3. Select **New Simple Volume → Next**, enter the size in MB, assign a drive letter (e.g. E:).
4. Choose **Format this volume**: file system **NTFS**, allocation unit default, volume label (e.g. DATA), tick *Quick Format* → Next → Finish.

Observation: a new drive E: (DATA) appears in File Explorer.

a2) Identify whether the disk is MBR or GPT

1. In Disk Management right-click **Disk 0** (the left box, not the partition) → **Properties → Volumes tab**.
2. Check **Partition style**: *Master Boot Record (MBR)* or *GUID Partition Table (GPT)*.

(Alternative: `diskpart` → `list disk` → a star (*) in the **Gpt** column means GPT.)

MBR	GPT
Max 2 TB, 4 primary partitions	Over 2 TB, up to 128 partitions
BIOS (legacy)	UEFI

a3) Shrink and extend a partition

- **Shrink:** right-click the volume (e.g. D:) → **Shrink Volume** → enter the amount to shrink in MB → **Shrink**. Unallocated space is created.
- **Extend:** right-click the volume that is next to the unallocated space → **Extend Volume** → Next → select the space → Finish. (Extend is greyed out if the free space is not adjacent.)

a4) Create a manual restore point, uninstall a program, roll back

1. **Enable protection:** search *Create a restore point* → System Properties → **System Protection** → select C: → **Configure → Turn on system protection**.
2. Click **Create**, type a name (e.g. "Before_Uninstall") → Create → "The restore point was created successfully".
3. **Uninstall a program:** Settings → Apps → Installed apps → select a program (e.g. a test application) → **Uninstall**.
4. **System Restore:** System Properties → **System Restore → Next** → choose "Before_Uninstall" → Next → Finish → Yes. The PC restarts.
5. **Observation:** after restart the program is back; the message "System Restore completed successfully" appears.

Result: partition work and System Restore were done successfully.

TASK b) Network configuration and troubleshooting commands (10 marks)

b1) Configure IP address, subnet mask, gateway and DNS

Procedure: Control Panel → Network and Sharing Centre → **Change adapter settings** → right-click **Ethernet → Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties** → *Use the following IP address*:

Setting	Example value
IP address	192.168.1.20

Subnet mask	255.255.255.0
Default gateway	192.168.1.1
Preferred DNS	8.8.8.8
Alternate DNS	8.8.4.4

Click **OK**. Verify with `ipconfig /all`. (Command-line method: `netsh interface ip set address "Ethernet" static 192.168.1.20 255.255.255.0 192.168.1.1`.)

b2) Test connectivity and troubleshoot

Command	Purpose	Observation (example)
`ipconfig /all`	Shows IP, mask, gateway, DNS, MAC	IP 192.168.1.20
`ping 192.168.1.1`	Tests connection to gateway	Reply, 0% loss, time <1 ms
`ping 8.8.8.8`	Tests Internet connectivity	Reply from 8.8.8.8
`ping www.google.com`	Tests DNS name resolution	Reply (name resolved to IP)
`tracert www.google.com`	Shows every router (hop) to the destination and delay	Hop list ending at destination
`netstat -an`	Shows open/listening ports and connections	e.g. 135, 445 LISTENING
`netstat -ano`	Same with the Process ID (PID)	Use Task Manager to find the PID
`ipconfig /flushdns`	Clears the DNS cache	"Successfully flushed the DNS Resolver Cache"

Result: IP configuration done; connectivity verified; open ports checked; DNS cache flushed.

TASK c) Firewall, folder sharing and VM / VPN (15 marks)

c1) Windows Firewall and network profile

1. Settings → Network & Internet → Ethernet → **Network profile type**: select **Private** (trusted office) or **Public** (untrusted).
2. Control Panel → **Windows Defender Firewall** → check it is **On** for Private and Public networks.
3. **Allow an app:** *Allow an app or feature through Windows Defender Firewall* → Change settings → tick the app (e.g. *File and Printer Sharing*) for **Private** only.
4. **Block a port:** *Advanced settings* → *Inbound Rules* → *New Rule* → *Port* → *TCP 23* → *Block the connection* → all profiles → name "Block Telnet" → Finish.
5. Test with ping/telnet from another PC to verify the rule.

c2) Share a folder with another system

1. Create a folder (e.g. `C:\Shared`). Right-click → **Properties → Sharing → Advanced Sharing**.
2. Tick **Share this folder** → **Permissions** → remove *Everyone*, add the required user → allow **Read** (or Change) → OK.
3. Network profile must be **Private** and *Network discovery* and *File and printer sharing* must be on.
4. Note the network path `\\PC1\Shared`. On the other computer, press **Win + R**, type `\\192.168.1.20\Shared` → Enter (enter username/password if asked).
5. **Observation:** the folder opens on the other computer; files can be copied according to the permission given.

c3) Install a virtual machine (hypervisor) *or* a VPN app

Option 1 – VM using VirtualBox:

1. Download and install **Oracle VirtualBox** (enable virtualisation in BIOS if required).
2. Click **New** → name (e.g. Ubuntu-Test) → select the ISO file → set RAM (2–4 GB), CPU (2) and virtual disk (20 GB) → Finish.
3. **Start** the VM → install the guest OS → set the network adapter to **NAT** (or Bridged).
4. Inside the VM, run `ip a` / `ipconfig`, `ping 8.8.8.8` and `tracert` / `traceroute`.

Option 2 – VPN app: install a trusted VPN app (e.g. ProtonVPN free), sign in, click **Connect**, and note the new IP address and country. Check with `ipconfig` and `tracert`; disconnect and compare.

c4) Basic network commands and network usage

- Run `ipconfig`, `ping`, `tracert`, `netstat`, `nslookup` inside/outside the VM or VPN.
- Note network usage: **Task Manager → Performance → Ethernet** (send/receive speed), **Resource Monitor → Network** (per-process usage), or Settings → Network & Internet → **Data usage**. Record the data sent/received in the observation book.

Result: firewall, folder sharing and VM/VPN were configured and tested successfully.

TASK d) Documentation (10 marks)

d1) E-waste disposal procedure for the old computers

1. **Inventory:** list the old computers (asset number, make, condition) and obtain management approval for disposal.
2. **Data security:** back up important data; wipe hard disks (secure erase) or physically destroy them.
3. **Segregation (sorting):** reusable (refurbish/donate), recyclable (metal, plastic, glass), hazardous (batteries, CRT, lead, mercury).
4. **Collection:** store in a safe, dry place; do not burn or throw in the dustbin.
5. **Hand over** to an **authorised e-waste recycler/collection centre** (as per CPCB / E-Waste (Management) Rules, 2022); take a receipt and disposal/recycling certificate.
6. **Recovery:** the recycler dismantles and recovers copper, gold, aluminium and plastic; hazardous parts are treated safely.
7. **Record:** update the stock register and keep documents for audit.

d2) Short lab setup plan (10 computers)

Area	Plan
Hardware purchase	Decide requirements (10 PCs with SSD, 8 GB RAM, monitors, UPS, switch, router, Wi-Fi AP); prepare the specification; get at least 3 quotations; compare price, warranty and service; place the purchase order; check items on delivery and enter in the stock register.
Installation	Plan layout and structured Cat6 cabling; install the OS and software from an image; configure IP, DNS, users and shared folders; test the network.
Electrical safety	Proper earthing, MCB/ELCB, surge protection, UPS, no overloaded sockets, good wiring, CO_2 fire extinguisher, regular inspection.
Cybersecurity	Firewall ON, antivirus with updates, Windows patches, strong passwords and limited user rights, regular backups, restricted Wi-Fi access, physical security and user awareness.

TASK e) Portfolio evaluation (5 marks)

Keep ready a neat **portfolio/observation book** containing:

- Date, aim, procedure and result for tasks a to d.
- Screenshots: Disk Management (new partition, shrink/extend), restore point, System Restore, IP configuration, ping/tracert/netstat/flushdns output, firewall rule, shared folder, VM/VPN.
- The e-waste procedure and lab setup plan, signed by the staff member.